Cross-Channel Relational Knowledge Distillation for Low-Electrode SSVEP Recognition

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Abstract. Brain–Computer Interfaces (BCIs) based on Steady-State Visual Evoked Potentials (SSVEP) typically rely on multiple occipital electrodes, which complicates wearable use. Knowledge Distillation (KD) could transfer accuracy from a multi-electrode teacher to a low-electrode student, but most KD methods assume that both models observe the same input. We propose Cross-Channel Relational Knowledge Distillation (CCKD), which trains the student to preserve the teacher's similarity structure among samples rather than its outputs or features. We evaluated CCKD with the FB-tCNN backbone on the Benchmark and BETA datasets, using five independent subject splits and eight seeds per split. CCKD improved three-electrode and one-electrode students in all four settings, by 0.71 to 1.19 percentage points on average (Holm-corrected $p \leq 0.029$). Response-based KD, a hybrid objective, FitNets, and attention transfer each reduced the mean accuracy in at least two settings, and CCKD outperformed FitNets for one-electrode students. CCKD leaves the inference cost unchanged and thus recovers part of the accuracy lost to electrode reduction, although the gains are modest and were established on one backbone.

Keywords: Brain–computer interface (BCI), Steady-state visual evoked potential (SSVEP), Knowledge distillation, Relational knowledge distillation, Cross-channel learning, Low-electrode EEG.

1 Introduction

Electroencephalography (EEG)-based Brain–Computer Interfaces (BCIs) enable direct communication between the human brain and external devices without requiring muscular activity. Among EEG-based BCI paradigms, Steady-State Visual Evoked Potential (SSVEP) has been widely adopted because of its high information transfer rate, short calibration time, and good stability. SSVEP has been applied to device control, assistive technologies, virtual reality, and mobile BCI systems [1,2].

Most SSVEP recognition methods use EEG signals from multiple occipital electrodes to exploit spatial information [3,4]. Many electrodes increase the preparation time and complicate wearable BCI systems. Edge computing platforms are also

constrained by the number of sensors. Therefore, SSVEP recognition with fewer electrodes at a comparable accuracy has become an important research direction [2,5,6].

Deep learning models now dominate SSVEP recognition because they learn directly from EEG signals without manual feature engineering [7,8,9,11,12,13]. Knowledge Distillation (KD) transfers knowledge from a teacher model to a student model, so that the student keeps accuracy at a lower cost [14,15,16,17,18]. However, most KD methods assume that the teacher and the student observe the same input space. The student can then mimic the teacher's output distribution or features directly. When the teacher uses nine electrodes and the student only one, this assumption no longer holds. The teacher should instead transfer knowledge that depends less on the electrode configuration. In EEG analysis, distillation has been applied across tasks, time windows, and recording sites [21,22,23], but not between models with different numbers of scalp electrodes.

This paper proposes Cross-Channel Relational Knowledge Distillation (CCKD) for low-electrode SSVEP recognition. CCKD builds the similarity structure among samples from the features of each model and trains the student to preserve the teacher's structure. CCKD is used only during training and does not change the inference cost of the student. Our contributions are as follows:

- We propose CCKD, a KD framework for a teacher and a student that use different EEG electrode configurations.
- On the Benchmark and BETA datasets, we show with five subject splits and eight seeds per split that CCKD improves three-electrode and one-electrode students without changing their inference cost.
- We compare CCKD with response-based, hybrid, feature-based (FitNets), and attention-based KD. Only CCKD gives a positive mean gain in all four settings. We also report results across window lengths, the information transfer rate, and the latency on a Raspberry Pi 4.

2 Related Work and Research Gap

2.1 Deep Learning for SSVEP Recognition

Canonical Correlation Analysis (CCA) and Filter Bank CCA (FBCCA) are widely used for SSVEP recognition because they need no training data and have a low computational cost [19,20]. Their performance, however, often drops for noisy signals or short time windows.

Deep learning models learn spatial and temporal features directly from EEG signals. Proposed architectures include EEGNet [7], Compact-CNN [8], ConvCA [10], FB-tCNN [9], a deep neural network by Guney et al. [11], and Transformer-based models [12]; a recent survey compares them in more detail [13]. These models outperform traditional methods on standard datasets but are developed with multiple occipital electrodes. ConvCA compresses multi-channel EEG into a single-channel representation

within the architecture. CCKD pursues a related goal through the training objective and leaves the student unchanged at inference.

2.2 Knowledge Distillation for EEG Analysis

KD was proposed to build compact models that keep the accuracy of a larger teacher [14]. Early methods used the teacher's output distribution as the target. Later methods distill intermediate features (FitNets) [15], attention maps [26], or relationships between samples [16,17,18]. In EEG analysis, KD has been applied to tasks such as emotion recognition [21]. Most of these methods assume that the teacher and the student use the same number of channels or the same feature space.

2.3 Distillation Across Different Inputs

Several studies extend KD to a teacher and a student that observe different inputs. Similarity-Preserving KD shows that relationships between samples reduce the dependence on the teacher's absolute representations [16]. In SSVEP, RMKD transfers knowledge from long to short time windows [22]. MESD transfers knowledge from scalp-EEG to ear-EEG by combining feature and response distillation [23]. Li et al. [25] transfer knowledge across subjects to reduce the calibration burden. These works cover the temporal, recording-site, and subject axes. No study has addressed transfer along the electrode-count axis within scalp EEG, that is, between models that observe the same site and subjects but a different number of channels. CCKD addresses this gap.

3 Proposed Method

3.1 Framework Overview

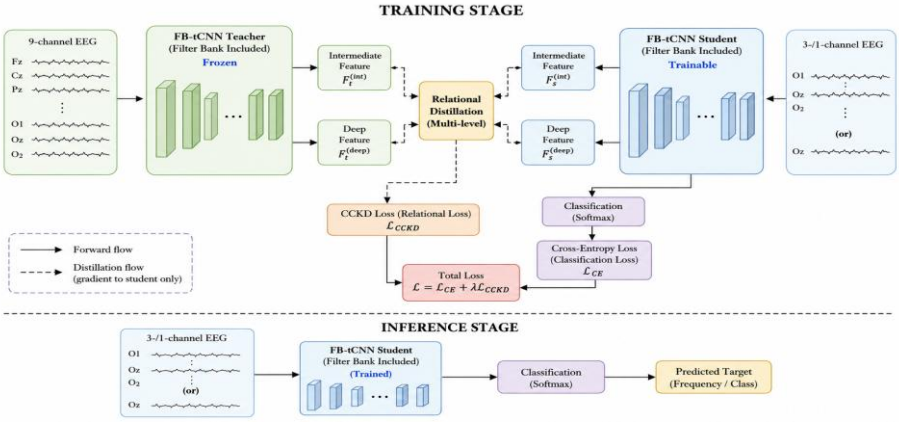


Fig. 1. Overall architecture of the proposed Cross-Channel Relational Knowledge Distillation (CCKD) framework for low-electrode SSVEP recognition.

Figure 1 gives an overview of CCKD. The teacher uses nine occipital electrodes (Pz, PO5, PO3, POz, PO4, PO6, O1, Oz, and O2). The student uses three electrodes (O1, Oz, O2) or one electrode (Oz). Both use the same backbone, so any difference in accuracy comes only from the electrode configuration. During training, the pretrained teacher is frozen. Each model extracts features at two representation levels. CCKD builds relational matrices from these features and trains the student to preserve the teacher's relational structure.

3.2 Multi-level Feature Extraction

CCKD distills from two layers of the backbone rather than from the final layer alone. For a mini-batch of N samples, the teacher and student features at layer l are $F_T^{(l)} \in \mathbb{R}^{N\times d_l^T}$ and $F_S^{(l)} \in \mathbb{R}^{N\times d_l^S}$. The Gram matrix $FF^\top$ is always $N \times N$, so CCKD does not require $d_l^T = d_l^S$. In this work, the two models share a backbone and $d_l^T = d_l^S$, but the formulation extends to heterogeneous pairs without modification. We use an intermediate layer with 736 dimensions and the final representation layer with 32 dimensions (Fig. 2). The first layer mainly carries local information, while the second carries high-level information.

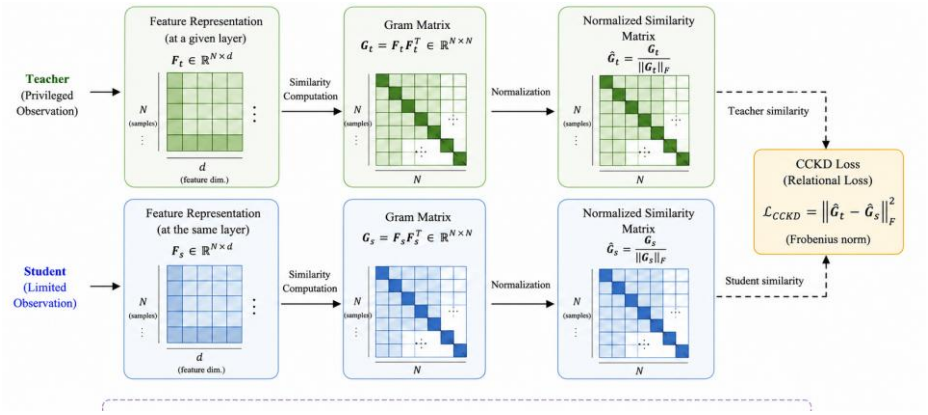


Fig. 2. Multi-level relational knowledge transfer between teacher and student feature representations.

3.3 Multi-level Relational Knowledge Distillation

The teacher and the student observe different electrode configurations, so their features cannot be matched sample by sample. CCKD instead transfers the relationships between samples [16]. At layer l, these relationships are represented by the Gram matrices

$$G_T^{(l)} = F_T^{(l)}(F_T^{(l)})^\top, \qquad G_S^{(l)} = F_S^{(l)}(F_S^{(l)})^\top, \tag{1}$$

where $G_T^{(l)}, G_S^{(l)} \in \mathbb{R}^{N\times N}$. Each element measures the similarity between a pair of samples, so the matrix describes the relational structure of the whole mini-batch. To remove the influence of feature magnitude, each row of a Gram matrix is normalized by its ℓ_2 norm:

$$\tilde{G}^{(l)}[i,:] = \frac{G^{(l)}[i,:]}{\|G^{(l)}[i,:]\|_2}, \qquad i = 1, \dots, N, \tag{2}$$

which is applied to both $G_T^{(l)}$ and $G_S^{(l)}$. Row i describes how sample i relates to the rest of the mini-batch. After normalization, the loss depends only on the relative similarity structure. The relational loss at layer l is the squared Frobenius distance between the normalized matrices:

$$\mathcal{L}_{\text{CCKD}}^{(l)} = \frac{1}{N^2}\left\|\tilde{G}_T^{(l)} - \tilde{G}_S^{(l)}\right\|_F^2. \tag{3}$$

3.4 Training

Algorithm 1: Training procedure of CCKD

Input: Pretrained teacher f_T (9 channels), student f_S (3 or 1 channel), training set $\mathcal{D}$, coefficient λ
Output: Trained student model f_S

Freeze all parameters of the teacher f_T
Initialize the student parameters θ_S
for *each epoch* **do**
  for *each mini-batch* $(X, y) \subset \mathcal{D}$ **do**
    Extract teacher channels X_T and student channels X_S from X
    $\{F_T^{(l)}\} \leftarrow f_T(X_T)$ `// teacher features, no gradient`
    $\{F_S^{(l)}\}, z_S \leftarrow f_S(X_S)$ `// student features and logits`
    $\mathcal{L}_{\text{CE}} \leftarrow \text{CrossEntropy}(z_S, y)$
    for $l \in \{1, 2\}$ **do**
      $G_T^{(l)} \leftarrow F_T^{(l)}(F_T^{(l)})^\top$; $G_S^{(l)} \leftarrow F_S^{(l)}(F_S^{(l)})^\top$
      Row-normalize: $\widetilde{G}_T^{(l)}[i,:] \leftarrow G_T^{(l)}[i,:]/\|G_T^{(l)}[i,:]\|_2$, same for $\widetilde{G}_S^{(l)}$
      $\mathcal{L}_{\text{CCKD}}^{(l)} \leftarrow \frac{1}{N^2}\left\|\widetilde{G}_T^{(l)} - \widetilde{G}_S^{(l)}\right\|_F^2$
    end
    $\mathcal{L}_{\text{CCKD}} \leftarrow \frac{1}{2}(\mathcal{L}_{\text{CCKD}}^{(1)} + \mathcal{L}_{\text{CCKD}}^{(2)})$; $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{CCKD}}$
    Update θ_S by minimizing $\mathcal{L}$
  end
end
return student model f_S

With $L = 2$ feature layers, the CCKD loss is the mean of the layer losses,

$$\mathcal{L}_{\text{CCKD}} = \frac{1}{L}\sum_{l=1}^{L}\mathcal{L}_{\text{CCKD}}^{(l)}, \tag{4}$$

which gives both levels equal weight. The student minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \mathcal{L}_{\mathrm{CCKD}}, \tag{5}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss and λ controls the relational term. We set $\lambda = 100$ in all experiments. Algorithm 1 summarizes the procedure. Only the student is updated. After training, the teacher is removed, and the student keeps its architecture, parameter count, and inference cost.

During training, CCKD adds one forward pass of the frozen teacher per mini-batch, without back-propagation through the teacher. It also computes four $N \times N$ Gram matrices. For $N = 64$, these matrices need about 6.3×10^6 multiply–accumulate operations per mini-batch ($2 \times 64^2 \times (736 + 32)$) and occupy 64 KB in single precision.

4 Experimental Setup

4.1 Datasets and Electrode Configurations

We use two public 40-target SSVEP datasets, Benchmark [3] and BETA [4], both sampled at 250 Hz (Table 1). Each subject split assigns disjoint subjects to the training, validation, and test sets. The validation subjects select the checkpoint, and the test subjects are evaluated once. To measure the dependence on the split, we repeat every experiment on five independent splits (split seeds 2026, 3026, 4026, 5026, and 6026).

Table 1. Summary of experimental settings.

Dataset	Subjects	Train/Val/Test	Splits	Window	Teacher	Student
Benchmark	35	21 / 4 / 10	5	1.0 s	9 channels	3 / 1 channel
BETA	70	42 / 8 / 20	5	1.0 s	9 channels	3 / 1 channel

The teacher uses the nine parieto-occipital electrodes common in SSVEP studies [20]. The students keep the electrodes over the primary visual cortex, where the SSVEP response is strongest: Oz for one channel and O1, Oz, O2 for three channels. Occipital sites are also used in wearable and single-channel SSVEP datasets [5,6]. We fix this choice a priori and do not search over electrode subsets.

4.2 Implementation Details

All models use FB-tCNN [9] with three filter-bank sub-bands. The signal window starts 0.14 s after stimulus onset to account for the visual latency. The two distillation layers of Section 3.2 are the output of the last temporal convolution block ($d_1 = 736$) and the final representation layer ($d_2 = 32$). The teacher is trained on nine-electrode data and then frozen. Students are trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}), cosine annealing, a batch size of $N = 64$, and 100 epochs. Each configuration is run with eight seeds (2026–2033) per split, and per-subject accuracies are averaged over seeds.

4.3 Comparison Methods and Evaluation Metrics

We compare CCKD with the nine-electrode teacher (upper reference) and a Student Baseline trained with cross-entropy only. We also compare four KD objectives on the same student: response-based KD [14] with $\tau = 4$, a hybrid of response-based KD and CCKD, FitNets [15], and attention transfer (AT) [26]. FitNets and AT act on the same two feature layers as CCKD and use the same weight $\lambda = 100$. This value gave the best result for both in a preliminary sweep over $\{0.1,1,10,100,1000\}$ on Benchmark with three electrodes.

We report the classification accuracy on the test subjects and the gain Δ over the Student Baseline in percentage points (pp). On the first split, we also report the number of improved test subjects and the Wilcoxon signed-rank test [24] over per-subject accuracies. Across the five splits, we report the mean and standard deviation (SD) of Δ and a one-sample t-test (p_t) of $H_0: \bar{\Delta} = 0$. Where several tests form one family, we apply the Holm–Bonferroni correction (p_{Holm}). We also report the information transfer rate (ITR) of Wolpaw et al. [27], with 40 targets and a gaze-shift time of 0.5 s.

5 Results and Discussion

5.1 Overall Performance

On the first split (2026), CCKD improved the Student Baseline in all four settings (Table 2). The gains were significant on BETA for three electrodes (+1.05 pp, $p = 0.012$) and one electrode (+1.06 pp, $p = 0.004$). On Benchmark, the gains of +0.45 and +0.59 pp did not reach significance with ten test subjects.

Across the five splits, the mean gain was positive in all four settings, and 19 of the 20 split-level gains were positive. All four gains remained significant after Holm correction over the four tests ($p_{\mathrm{Holm}} \leq 0.029$). The three-electrode gain on Benchmark averaged +1.00 pp, more than twice its value on the first split. A single split with ten test subjects can therefore underestimate the effect.

Table 2. Accuracy of CCKD and the Student Baseline (FB-tCNN, 1.0 s window). Left: first split (seed 2026, 8 seeds), with improved test subjects and the Wilcoxon p. Right: gain Δ over five splits (mean $\pm$ SD), positive splits, p_t, and p_{Holm} over the four tests. Teacher accuracy on split 2026: 65.69% (Benchmark), 53.91% (BETA).

		Split 2026					Five splits			
Dataset	Ch.	Base (%)	CCKD (%)	Δ (pp)	Impr.	p	$\bar{\Delta} \pm$ SD	Pos.	p_t	p_{Holm}
Benchmark	3	50.99	51.44	+0.45	6/10	0.375	**+1.00** $\pm$ 0.40	5/5	0.005	**0.015**
	1	44.55	45.14	+0.59	7/10	0.193	**+0.71** $\pm$ 0.35	5/5	0.010	**0.021**
BETA	3	38.41	39.46	**+1.05**	16/20	**0.012**	**+1.19** $\pm$ 0.21	5/5	0.0002	**0.001**
	1	34.62	35.68	**+1.06**	15/20	**0.004**	**+0.72** $\pm$ 0.48	4/5	0.029	**0.029**

5.2 Comparison with Other Distillation Objectives

Table 3 compares the KD objectives over the same five splits. Response-based KD and the hybrid objective did not improve the students consistently. Both reduced the one-electrode accuracy on BETA in all five splits (-1.16 and -0.84 pp, $p_t = 0.009$ and 0.036). Adding the response term to CCKD thus removed the benefit of the relational term.

Table 3. Gain Δ (pp) over the Student Baseline, mean over five splits (positive splits in parentheses). Bold: $p_t < 0.05$ (one-sample t-test, uncorrected). FB-tCNN, 1.0 s window; response KD uses $\tau = 4$.

Objective	Bench. 3C	Bench. 1C	BETA 3C	BETA 1C
Response KD [14]	+0.49 (4/5)	−0.76 (2/5)	+0.17 (2/5)	$\mathbf{-1.16}$ (0/5)
Hybrid (response + CCKD)	+0.57 (4/5)	−0.48 (2/5)	+0.37 (3/5)	$\mathbf{-0.84}$ (0/5)
FitNets [15]	**+1.09** (5/5)	−0.10 (2/5)	**+0.95** (5/5)	−0.44 (1/5)
Attention transfer [26]	**+0.77** (5/5)	−0.30 (1/5)	−0.66 (2/5)	−2.76 (0/5)
CCKD (ours)	**+1.00** (5/5)	**+0.71** (5/5)	**+1.19** (5/5)	**+0.72** (4/5)

The feature-based objectives depended on the number of student electrodes. With three electrodes, FitNets matched CCKD, with no difference in the paired t-test over splits (Benchmark $p = 0.39$, BETA $p = 0.14$). AT improved the three-electrode student on Benchmark but not on BETA. With one electrode, the mean gains of both objectives were zero or negative, whereas CCKD remained positive. CCKD exceeded FitNets in all five splits on both datasets (paired $p = 0.005$ and 0.001). It also exceeded AT in all five splits (paired $p = 0.004$ on Benchmark and 0.058 on BETA). CCKD was thus the only objective with a positive mean gain in all four settings, but it did not outperform FitNets with three electrodes.

5.3 Window Length and Information Transfer Rate

Table 4 repeats the five-split evaluation for windows of 0.5 to 2.0 s. For three electrodes, the gain was positive in all five splits for every window of at least 1.0 s on both datasets. After Holm correction over the 16 tests, three settings remained significant: BETA at 1.0 and 2.0 s and Benchmark at 2.0 s ($p_{\text{Holm}} \leq 0.011$). For one electrode, the gains were smaller, and none survived the correction. At 0.5 s, the effect was weak on both datasets.

CCKD raised the ITR in every setting. The largest absolute gains occurred at 1.0 to 1.5 s, for example from 79.9 to 82.1 bits/min for three electrodes on Benchmark at 1.0 s. On Benchmark, the three-electrode ITR peaked at 1.0 s, while the one-electrode ITR still rose up to 2.0 s. On BETA, both ITRs were highest at 2.0 s. The 3.0 s trials of BETA limit the window to about 2.36 s after the latency offset, so longer windows were not tested.

Table 4. CCKD gain over five splits by window length (mean Δ in pp, p_t in parentheses), and ITR in bits/min of the Student Baseline → CCKD, computed from the five-split mean accuracy. Bold: $p_t < 0.05$.

Window	Dataset	Δ 3C	Δ 1C	ITR 3C	ITR 1C
0.5 s	Benchmark	+0.37 (0.27)	+0.04 (0.87)	35.4 → 36.2	30.7 → 30.8
	BETA	**+0.68** (0.023)	+0.25 (0.068)	22.4 → 23.7	17.8 → 18.2
1.0 s	Benchmark	**+1.00** (0.005)	**+0.71** (0.010)	79.9 → 82.1	65.1 → 66.6
	BETA	**+1.19** (0.0002)	**+0.72** (0.029)	42.7 → 44.9	35.9 → 37.2
1.5 s	Benchmark	**+1.18** (0.020)	+0.53 (0.076)	79.2 → 81.5	70.1 → 71.0
	BETA	**+1.32** (0.004)	+0.45 (0.150)	49.0 → 51.1	44.2 → 44.9
2.0 s	Benchmark	**+0.72** (0.0008)	+0.20 (0.28)	78.2 → 79.4	71.2 → 71.5
	BETA	**+1.10** (0.0001)	**+0.52** (0.012)	49.3 → 50.8	46.1 → 46.8

5.4 Deployment on a Raspberry Pi 4

To check the edge deployment, we measured the inference latency on a Raspberry Pi 4 Model B (Cortex-A72). We used a 1.0 s window and a batch size of one. We measured the filter-bank preprocessing and the forward pass separately, each with 20 warm-up and 200 timed iterations, and repeated the process seven times. With one CPU thread, the summed latency was 17.24 ± 0.05 ms for the teacher, 16.82 ± 0.23 ms for the three-electrode student, and 16.53 ± 0.21 ms for the one-electrode student (mean $\pm$ 95% confidence interval). With four threads, it was 13.2 to 14.0 ms. Adding this latency to the trial time lowered the ITR by about 1.1%, and the peak memory was 298 MB.

The three models had almost the same latency and parameter count (90,536, 90,440, and 90,408 parameters). The first convolution of FB-tCNN collapses the channel dimension, so fewer electrodes barely reduce the computation. The benefit of fewer electrodes therefore lies in simpler sensing hardware and shorter preparation, not in faster inference.

5.5 Discussion

Taken together, the comparisons indicate that the type of transferred knowledge decides whether distillation helps when the teacher and the student observe different electrode sets. A nine-electrode teacher and a one-electrode student can produce different posteriors for the same trial. The teacher's softened outputs then encode a confidence that the student cannot reproduce from its input. This mismatch may explain why response-based and hybrid KD reduced the one-electrode accuracy on BETA. Feature imitation faces a related problem, because the absolute features of the two models need not align. Relational supervision constrains only which samples should be similar, a target that remains defined when the student observes much less of the signal. The one-electrode results, where CCKD exceeded FitNets on both datasets, are consistent with this account, although we did not test the mechanism directly.

These findings extend similarity-preserving KD [16], introduced for model compression with a shared input, to students that observe fewer input channels. They also

qualify the use of response terms in cross-input SSVEP distillation, which MESD combines with feature distillation for scalp-to-ear transfer [23]. For electrode reduction on the scalp, adding a response term to CCKD removed its benefit.

The gains are modest relative to the cost of removing electrodes. Over five splits at 1.0 s, the teacher exceeded the CCKD students by 9.6 to 17.0 pp. CCKD closed about 4 to 9% of the gap between the Student Baseline and the teacher, so it reduces this cost but does not remove it.

Four boundaries limit the scope of these claims, and each points to a specific next test. First, all results use FB-tCNN, and preliminary runs with other backbones suggest that the gain depends on the backbone and the dataset, which limits the claims to this architecture. Second, the electrode subsets were fixed a priori, so the gain may differ for other montages. Third, $\lambda = 100$ was not selected on validation data. The positive gains on the four additional splits reduce, but do not exclude, the risk of an optimistic choice. Fourth, within one seed, the baseline and KD runs share the random-number state of one process. The baseline also uses mixed precision, while the KD runs use single precision. The paired design and the consistency across splits limit, but do not exclude, an effect of these differences on the estimated gains.

6 Conclusion

This study proposed Cross-Channel Relational Knowledge Distillation (CCKD) to transfer knowledge from a multi-electrode teacher to a low-electrode student for SSVEP recognition. CCKD transfers the similarity structure among samples rather than outputs or absolute features. It therefore works across electrode configurations and leaves the architecture and inference cost of the student unchanged.

With FB-tCNN, five subject splits, and eight seeds per split, CCKD improved three-electrode and one-electrode students on Benchmark and BETA. The mean gains were 0.71 to 1.19 pp, and all four were significant after Holm correction. Response-based, hybrid, feature-based, and attention-based KD each gave a negative mean gain in at least two settings. CCKD was the only objective with a positive mean gain in all four settings, and it outperformed FitNets with one electrode on both datasets. The full inference pipeline runs in under 18 ms on a Raspberry Pi 4, and CCKD adds no inference cost to the student. Future work will address the boundaries above by identifying which backbone properties determine the gain, measuring its sensitivity to the electrode montage, and selecting λ on validation data.

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